AI for Good Sandbox Hackathon – PrismAI – Prism: agentic AI approval and assurance for public services

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Github link: https://github.com/devinwarp/prismai_UN_A4G. Licence: AGPL-3.0 (a commercial licence is available from the copyright holders).

1. Introduction

On 23 April 2026 the UAE announced a plan to move 50% of government sectors, services and operations to agentic AI within two years (about April 2028; assumed, as no date was published). On 18 May 2026 the Cabinet adopted a framework to transform at least 50 per cent of federal services and operations into agentic AI systems, and asked each entity to "set targets and establish assessment indicators". The sources we ingested leave three things open: what counts towards the 50 per cent, how much autonomy an agent may be given, and who may approve an agent going live.

Prism is an agentic approval and assurance layer for that gap. A policy is configured (jurisdiction, target %, deadline, counting rule). Use cases are imported and tagged to the four quadrants of the Digital Dubai AI Integration Matrix. Role agents (innovator, policy and legal adviser, security, privacy, compliance, frontline, finance, coordinator) examine each case against a knowledge base built with ITU's reference pipeline. A deterministic policy engine, not a model, issues the verdict, and each entity can set its own human-in-the-loop rules (below). Every step is written to a hash-chained trace log, and a Sijil panel supplies run-level evidence. A federal dashboard reports deployed % beside compliant %; a run viewer shows the step timeline, gate matrix and hash-chain status. All entities and data are fictitious.

This report maps the solution to the ITU-T Y.3172 pipeline and to the 13 dimensions and 6 factors of the AI Readiness Report 2.0, runs three staged error scenarios, and lists the policy and standards gaps the mapping exposed.

2. Description of the use case

Domain: public services. Primary case UC02, trade-licence renewal and permit pre-check at Marsa Public Services Authority (fictitious): an external agent pre-checks a renewal and drafts the customer message. It sits in the Matrix's External Agents quadrant and is client-facing, so the entity rule is that an officer gives final approval before any message reaches a customer. The federal view covers 20 use cases across five fictitious entities (8 internal agents, 5 external agents, 3 internal RAG, 4 external RAG).

- **Counting rule.** The 50% target has no published metric. Prism makes the rule explicit and editable (default: a use case counts when its verdict is GO_AUTONOMOUS or GO_SUPERVISED) and reports compliant % beside deployed %, so speed cannot hide a safety failure.

- **Autonomy levels.** A five-step verdict scale (GO_AUTONOMOUS, GO_SUPERVISED, ASSISTANT_ONLY, NO_GO, ESCALATE) with a hard cap: a client-facing use case never exceeds GO_SUPERVISED.

- **Human in the loop.** Per-entity rules, edited on the Federal page (src/lib/un/hitl.ts): client-facing output needs a person's final approval; decisions affecting rights or fees need a human decision; irreversible actions need human approval. All three default to on, an entity can switch one off, and an

applicable rule declared unmet is a hard failure of the human-approval gate (verdict NO_GO), never averaged. The rules load into runs from the entity configuration.

• **Accountability.** Each hard gate has one owning role agent; only the owner can fail or pass it, and only the applicant can declare a fix. The verdict is computed by code from gate states, never averaged.

• **Auditability.** Every agent message, every rejected message and every verdict is a hash-chained record with quoted citations, and a run can be replayed from its log. Each trace record is hashed together with the previous record's hash, and the hash covers every field of the record, including its timestamp, the exact agent message, the evidence and knowledge-base citations, the proposal and disposition, the trace ids a verdict rests on, any HITL breach and the human signature. Editing any record, or deleting, inserting or reordering records, breaks the chain. A reader can check it offline with verify_log, which names the first broken record; dropping the last records is detected by comparing with the stored head hash. Separately, every change to a use case's stage, verdict or compliant flag, to the policy and to the human-oversight rules is written to an append-only, hash-chained change log with the old and new value and a UTC timestamp, verified by a database function that names the first broken row. The change log records the database role, not the person; who signed a verdict is on the use case and in the run trace. History starts when the log was installed.

3. Use case requirement

Requirements the solution is checked against (R1 and R3 to R7 have automated tests in the repository (113 unit tests pass with node --test); R2 is checked by running the build):

R1 Policy: target %, deadline and counting rule are configurable; deployed % and compliant % are reported separately.

R2 Knowledge base: every source carries its URL, a SHA-256 and, where the source states one, its date; a query returns cited chunks from UAE/Dubai and global sources; copyrighted sources (the AI Charter, all 12 principles; the UAE AI Strategy 2031; the PDPL, official text; the Dubai AI Ethics Guidelines) are held as short attributed excerpts with their URLs, not full text; the base is rebuilt by one script and vectorised with ITU's embedding function and model (nomic-embed-text, 768 dimensions, kb/vectors.json); the app currently searches it by keyword (BM25), and hybrid keyword-plus-vector search is built but switched off in the app.

R3 Agents: at least 5 role agents active; each sees only the events routed to it; typed messages (PROPOSE, CHALLENGE, EVIDENCE_REQUEST, EVIDENCE, POSITION_UPDATE, ESCALATE, VERDICT).

R4 Policy engine: five hard gates (approved infrastructure, audit log, lawful data, human final approval, AI disclosure) plus two metric gates added for S2, hallucination rate and fee accuracy; a message with no evidence, or citing a chunk that does not exist, is rejected.

R5 Traceability: verdicts list the records they rest on; the verifier detects edits, deletions, insertions and reordering.

R6 Evidence: at least 30 Sijil runs with accuracy ≥ 95%, hallucination ≤ 1%, goal alignment ≥ 90%, and zero rejects on hard criteria (the three rates are simulated here).

R7 Staged events: released one step at a time and unevenly distributed, never all at once.

4. **Mapping with ITU-T Y.3172, the AI Readiness Report 2.0, and domain-specific policies**

Report 2.0 factors: Data (knowledge base, use cases); Research (Singapore, EU and NIST benchmarks); Deployment support (infrastructure gate); Standards (Y.3172 pipeline, Matrix quadrants, verdict scale); Open

source and code (GitHub, ITU parsing and chunking functions reused; HTML cleaning is ours); Sandbox environments (scenario engine, simulated Sijil).

Y.3172 node	Prism use case	Readiness 2.0 Report (dimension)	Documents in the knowledge base
SRC	22 sources (331 chunks) pinned by SHA-256; 20-use-case CSV; staged scenario events; Sijil telemetry (simulated)	1 Data/Model Marketplace; 4 Contextualization and Regional Impact	UAE Cabinet, 23 April and 18 May 2026 (50% target, Phase One service categories); Digital Dubai AI Integration Matrix launch; UAE AI Strategy 2031; Dubai AI Ethics 2019 (excerpts)
C	ITU reference ingestion code (parse, chunk, embed); CSV import with line-level errors; simulated and optional live Sijil adapter	13 Digital Infrastructure	National Cyber Security Policy for AI (UAE); ITU AI Readiness reference repository
PP	Chunks with jurisdiction, instrument type, binding flag and date; Matrix-quadrant tagging; CSV validation	3 Cross-Domain Correlation Analysis	ITU reference chunking; kb/sources.json schema
M	Eight role agents (model-driven, with a deterministic mode) plus BM25 retrieval over the base	5 Level of Integration of AI in Workflows; 6 Human Interface; 8 Collaboration with AI	AI Integration Matrix (internal/external, agent/RAG); UAE AI Charter (human-machine ties); NIST AI RMF 1.0
P	Policy engine, the Y.3172 P node ("enables the application of policies to the output of the model node"): five hard gates, per-entity HITL rules, verdict scale, client-facing cap, citation gate	10 AI & Policies; 7 Strategy Alignment	UAE AI Charter (human oversight, transparency); Singapore agentic AI framework (override, intercept or review agent actions); EU AI Act Annex III (access to essential public services)
D	Federal dashboard; trace viewer; auto-filed pilot reports (Markdown, hashed)	7 Strategy Alignment; 12 Granular Priorities	UAE Cabinet (entity targets and assessment indicators)
SINK	Hash-chained trace log and filed reports; compliant flag read by the dashboard	9 Impacts of Humans in AI Integration; 10 AI & Policies	UAE AI Charter (accountability); NIST AI RMF (govern, measure)
MLFO / ML intent	Coordinator agent and scenario engine; the policy configuration and each use case's requested autonomy are the intent	7 Strategy Alignment	UAE Cabinet (50% target, targets and indicators per entity); Singapore framework (lifecycle accountability)
ML sandbox	Scenarios S1, S2 and S3 and the simulated Sijil adapter run on synthetic data only; S3 works on a copy of the use cases	9, 10, 11 (sandbox factor); 11 AI for Inclusion not yet addressed	Dubai Future Foundation AI Sandbox and Agentic AI accelerator; ITU AI4G sandbox summary

5. Error scenarios.

S1, UC02 (gate engine). Step-1 (T0): Marsa requests GO_AUTONOMOUS for a client-facing case. Baseline results go to the innovator, frontline and policy adviser; audit-log, lawful-data and disclosure evidence go to security, privacy and compliance. Hosting and human-approval gates have no evidence. Verdict ESCALATE (autonomy above the client-facing cap; two gates unevidenced).

Step-2 (T1): Only the security agent learns that the model endpoint is not on DESC-certified infrastructure. It raises a CHALLENGE; verdict NO_GO. No other agent sees this event.

Step-3 (T2): Only the compliance agent learns the agent sends customer messages without officer release. A second gate fails; NO_GO.

Step-4 (T3): The innovator declares both fixes. Declared but untested: ASSISTANT_ONLY (the agent prepares, a person acts).

Step-5 (T4): Re-tests pass. Verdict GO_SUPERVISED, the ceiling for a client-facing case, never autonomous.

Outcome. The verdict path matched the expected path at every step (ESCALATE, NO_GO, NO_GO, ASSISTANT_ONLY, GO_SUPERVISED). The run wrote 24 trace records from 7 active agents; the verifier passes, and changing one claim in record 4 fails verification at record 4. At T2 the compliance agent cites the Singapore framework (as summarised by Baker McKenzie): "human oversight mechanisms that can effectively override, intercept or review agentic AI actions." At T1 the security agent cites the DESC Information Security Regulation (a short excerpt on cloud security controls, desc-isr-excerpts#0) for the hosting gate.

S2, hallucination alert. UC01, voice contact centre (GO_SUPERVISED pilot, client-facing). T0: all five gates pass; GO_SUPERVISED. T1: only the compliance agent sees a Sijil alert (hallucination rate 6.1% over 3 runs against a 1% threshold) and the owner contains it at once by restricting fee answers to the fee table; fix declared, not re-tested; ASSISTANT_ONLY. T2: only the frontline agent sees a complaint that the assistant quoted a licence fee that does not exist; compliance links it to the T1 alert and adds the fee_accuracy gate, which fails; NO_GO. T3: only the policy adviser sees the requirement to tell users when AI-generated information may be inaccurate; the disclosure gate is re-tested against it and fails; two open failures, escalated to a person; NO_GO. T4: the owner declares fixes for both gates; ASSISTANT_ONLY. T5: all re-tests pass; GO_SUPERVISED, with the hallucination and fee-accuracy gates now permanent. The expected path is GO_SUPERVISED, ASSISTANT_ONLY, NO_GO, NO_GO, ASSISTANT_ONLY, GO_SUPERVISED; the Sijil figures are simulated scenario inputs.

S3, federal cascade. Policy set to 50% over the 20 use cases, on a sandbox copy. T0: 0 of 20 deployed (target 10). T1: entities report 6 deployed (30%, gap 4). T2: a Sijil attestation fails 2 of the 6, so compliant is 4 of 20 (20%). T3 (simulated policy update): a new Phase One service category is added, and the four transport use cases are flagged for re-scoring. The dashboard reads 30% deployed but 20% compliant: reporting both is the point. Live status (the deployed app, not the scenario): 2 of 20 use cases are deployed and 2 of 20 are compliant, UC01 and UC02, both signed by a person and attested, against a 50% target of 10 use cases. That is 10%, a gap of 8. Prism reports the gap; the target is not met.

6. Policy gaps and standards gaps derived from the mapping exercise

Gap	What the ingested sources say	Global reference	Prism response
Unclear responsibilities: who approves going live	No source ingested names who approves an agent. Each entity is asked to set targets and assessment indicators (uae-cabinet-2026-05-18#4); results are assessed per entity (uae-cabinet-agentic-50#1).	Singapore: humans made meaningfully accountable (sg-agentic-summary#1, law-firm summary; PDF not ingested). NIST AI RMF 1.0: accountability relates to all other trust characteristics (nist-ai-rmf#3).	One owning agent per gate; applicant may only declare fixes; a named person signs the final verdict

Standards: what counts as deployed, and agent quality metrics	Cabinet: at least 50 per cent of services and operations, with entity-set indicators (uae-cabinet-agentic-50#2). No metric for hallucination or goal alignment in the UAE sources ingested.	NIST AI RMF: Measure function (nist-ai-rmf#51). ITU AI Readiness 2025: detecting hallucination "could be considered" (itu-ai-ready-2025#28).	Explicit, editable counting rule; deployed % and compliant % side by side; Sijil metrics per criterion (simulated here)
Technology: applying policy to agent output before it acts	The Dubai framework requires an audit trail of agent actions (dubai-ai-matrix-excerpts#0). No source ingested requires policy checks on an agent's output before it acts.	ITU-T Y.3172: the P (policy) node applies policies to the model's output (itu-y3172-standard#33); the ML sandbox tests models before they touch the live network (itu-y3172-standard#10).	Policy engine is the P node: hard gates, client-facing cap, HITL rules; scenarios run as a sandbox on synthetic data
Legislative: citizen recourse and autonomy levels	April release: agents may act without human intervention; the Charter states human oversight as a principle (uae-ai-charter#2). The PDPL lets a person object to an automated decision (uae-data-protection-laws#10), ask for human review (#11) and complain to the Bureau (#14), but it does not apply to government data or governmental entities (#2). No recourse route against a federal agent or autonomy levels found.	NIST: processes to report problems and appeal outcomes (nist-ai-rmf#51). EU AI Act Annex III: access to essential public services is high-risk (eu-ai-act-annex3#2). Art. 86 not ingested.	Verdict scale with client-facing cap; human final approval as hard gate; S2 turns a complaint into a gate; no recourse workflow
Coordination: federal and Dubai instruments, hosting evidence	Federal data-sharing policy: collect once, use across entities (uae-cabinet-agentic-50#4). Dubai has its own matrix and the DESC Information Security Regulation, issued under Dubai Law 11 of 2014 (desc-isr-excerpts#5). The sources do not say how federal and Dubai rules relate for hosting.	ITU AI4G sandbox vision: a network of sandboxes across countries on one Y.3172 pipeline (itu-sandbox-summary#0). Second example pending: UN and G7 PDFs not ingested.	Hosting gate cites a short ISR excerpt on cloud security (desc-isr-excerpts#0); the full text is not redistributed; how the ISR applies to a federal entity is not stated
Conflicts between objectives: speed target versus assurance	Two-year timeline and performance assessment of entities (uae-cabinet-agentic-50#1); no stated weighting between speed and verified safety.	NIST AI RMF: trustworthiness characteristics involve tradeoffs that depend on context (nist-ai-rmf#22, #23); tradeoffs between societal values should be discussed (nist-ai-rmf#20).	Deployed % beside compliant % so speed is never reported without assurance; target and counting rule are configuration

Limits. Sijil is simulated by default and labelled so. The optional live adapter takes run counts, policy decisions, approval records and the signed hash-chained ledger verification from a Sijil node. Sijil does not measure accuracy, hallucination or goal alignment, so these stay simulated and labelled. The adapter is tested only with a mocked fetch, not yet against a real node. Agents were evaluated in deterministic mode; model-driven mode is built but not yet scored. Retrieval is lexical, so citation relevance is uneven. Retrieval was measured on 30 questions over the 331-chunk, 22-source knowledge base (kb/eval/RESULTS.md). The keyword search the app uses (BM25) reaches recall@5 of 0.87, hybrid keyword-plus-vector search 0.83 and vector-only search 0.50; on 14 held-out questions the figures are 0.86, 1.00 and 0.71. The plan's hybrid target of 0.85 is not met. Vector search is switched off in the live app and BM25 is the default; we do not claim vector search is live. The excerpt-chunk vectors are approximate, not the model's output, so vector and hybrid figures are indicative.